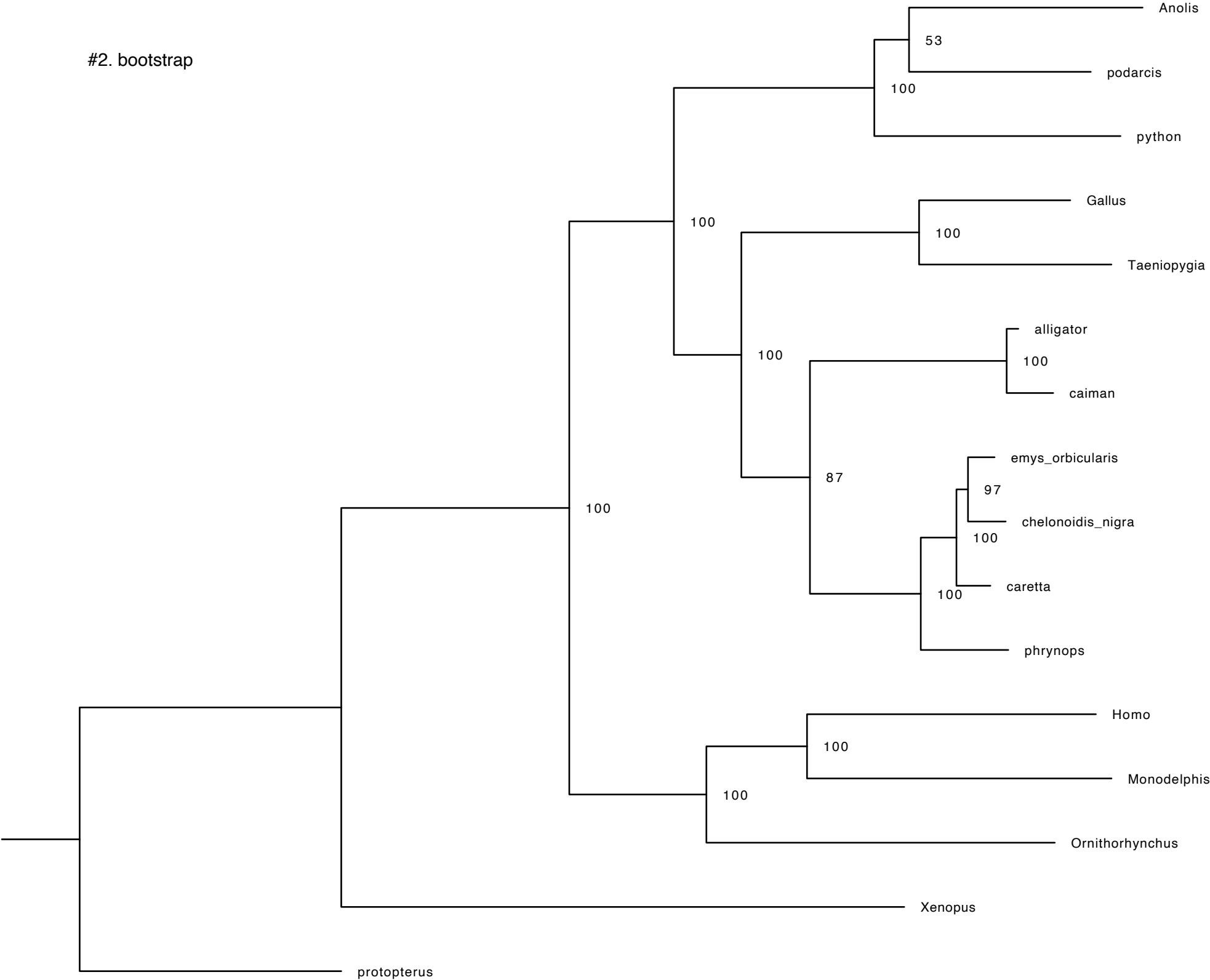
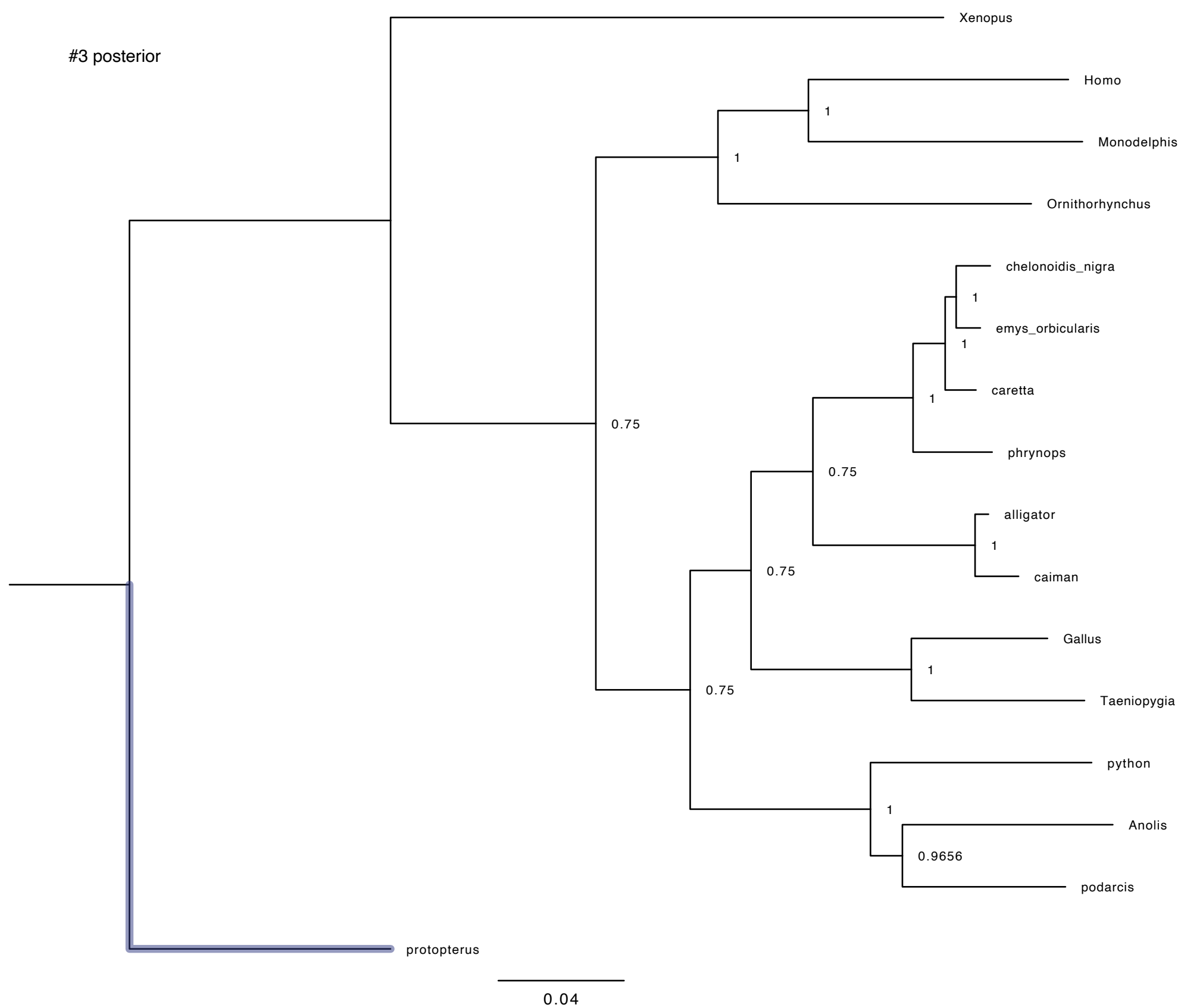


#2. bootstrap



0.03

#3 posterior



#4. When looking at the bootstrap proportions and the posteriors respectively of the two trees above, we do see that some of the estimates in the nodes are different from each other (while for some of the other nodes, the estimates are the same, ex: Gallus and Taeniopygia both have an estimate of 1/100. The evolutionary relationships shown in both trees are the same, however.

We ended up with different proportions/probabilities from using the bootstrap method vs. the MCMC method because the two methods sample in different ways to get their results. Bootstrapping relies on generating pseudoreplicates by sampling with replacement from the original dataset and then creating a distribution from those pseudoreplicates. The MCMC method, on the other hand, generates random samples/values and will accept said value if it's close enough to a distribution that's been predetermined.